

Probability. ①

Random Variable :-

A real variable X whose value is determined by the outcome of a random experiment, is called a random variable or a stochastic variable; i.e. a random variable is often described as a variable whose values are determined by chance.

eg ① Tossing of a coin twice

Sample space $S = \{ \underset{S_1}{HH}, \underset{S_2}{HT}, \underset{S_3}{TH}, \underset{S_4}{TT} \}$.

Define a fn $X: S \rightarrow \mathbb{R}$ by $X(S) = \text{No. of heads}$ then $X(S_1) = 2$, $X(S_2) = 1$, $X(S_3) = 1$
 $X(S_4) = 0$

$\therefore \text{Range} = \{2, 1, 0\} \Rightarrow \text{Random variable}$
 $X(S) = \{0, 1, 2\}$.

eg ② Throw of pair of dice

Sample space $S = 6^2 = 36$

$= \{ \begin{array}{lll} (1,1), (1,2), (1,3) & \dots & (1,6) \\ (2,1), (2,2), (2,3) & \dots & (2,6) \\ \vdots & & \vdots \\ (6,1), (6,2), (6,3) & \dots & (6,6) \end{array} \}$

(2)

Define a fn $X: S \rightarrow R$ by

$X(S)$ = Sum of two dice ie

$$X(i, j) = i + j \quad \forall (i, j) \in S$$

ie $X(S) = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$.

Types of Random Variable :-

They are of two types

Discrete Random variable

Continuous Random variable

Discrete Random variable :-

A random variable X which takes only a finite no. of discrete values in an interval of domain is called a Discrete Random variable.

OR

If the Random variable takes the value only in the set $\{0, 1, 2, \dots, n\}$ is called discrete random variable.

(3)

- eg 1) the throw of a dice,
2) Tossing of a coin.

Continuous Random variable :-

A random variable X which can take values continuously i.e. which takes all possible values in a given interval.

- eg 1) the height, age
2) Temp & time etc.

* ie if a random variable takes a finite set of values, it is called a discrete variable and if it assumes an infinite no. of uncountable values it is called a continuous variable.

Discrete Probability distribution :-

It is the set of its possible values together with their respective probabilities.

Let X be a discrete random variable with possible outcomes $x_1, x_2, x_3, \dots$

then their probabilities $P_i = P(X=x_i) = P(x_i)$

$$P_i = P(X=x_i) = P(x_i) \text{ for } i=1, 2, 3, \dots$$

where i) $P(x_i) \geq 0$ $\forall$ values of i

$$\text{ii) } \sum P(x_i) = 1.$$

then the set of values x_i with their probabilities P_i constitute a discrete probability distribution of the discrete variable X .

eg Tossing a coin two times.

Sample space = $\{HH, HT, TH, TT\}$

x_i	0	1	2
$P(x_i)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

$X(s) = \text{no. of heads} = \{0, 1, 2\}$

so here $\sum P(x_i) = 1$

$$\text{i.e. } \frac{1}{4} + \frac{2}{4} + \frac{1}{4} = \frac{4}{4} = 1 \text{ and } P(x_i) \geq 0.$$

Sample Point	X	X = no. of Heads.
T T T	$X(TTT)$	0
H T T, T H T, T T H	$X(HTT), X(THT), X(TTH)$	1
T H H, H T H, H H T	$X(THH), X(HTH), X(HHT)$	2
H H H	$X(HHH)$	3

(S)

$$A = \{0, 1, 2, 3\}$$

Probability Mass function :-

let- $x_1, x_2, x_3, \dots, x_n$ be the values of a discrete random variable X and let- $p_1, p_2, p_3, \dots, p_n$ ($p_i > 0, i = 1, 2, 3, \dots, n$) be the corresponding probabilities.

that is $P(X = x_i) = p(x_i) = p_i$.

A function $p(x)$ defined by

$$P(X = x) = p(x) = \begin{cases} p(x_i) \text{ or } p_i, & i = 1, 2, 3, \dots \\ 0 & , \text{ otherwise} \end{cases}$$

is called the Probability mass fn
of the discrete random variable X
where

$$i) p(x_i) \geq 0$$

$$ii) \sum p(x_i) = 1$$

Example Suppose a coin is tossed
three times. then the distribution of
the no. of heads is

$X = x :$	0	1	2	3
$P(x) :$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

here all $p(x_i) \geq 0$ and $\sum p(x_i) = 1$

$$\text{ie } \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{8}{8} = 1.$$

Mean and Variance of random variables :-

Let $X : x_1 \quad x_2 \quad \dots \quad x_n$

$P(X) : p_1 \quad p_2 \quad \dots \quad p_n$

be a discrete Probability
distribution.

$$\begin{aligned} \text{Mean} = \mu &= \frac{\sum p_i x_i}{\sum p_i} \\ &= \sum p_i x_i \quad \left\{ \text{since } \sum p_i = 1 \right\} \end{aligned}$$

$$\text{Variance } \sigma^2 = \sum p_i x_i^2 - \mu^2$$

example (1)

The Probability density fn of a Variate X is

$X:$ 0 1 2 3 4 5 6

$P(X):$ k $3k$ $5k$ $7k$ $9k$ $11k$ $13k$

i) Find $P(X < 4)$, $P(X > 5)$,
 $P(3 < X \leq 6)$.

ii) what will be the minimum value of k so that $P(X \leq 2) > 3$

Sol i) if X is a random variable then

$$\sum_{i=0}^6 P(x_i) = 1$$

i.e. $k + 3k + 5k + 7k + 9k + 11k + 13k = 1$

$$\Rightarrow 49k = 1 \Rightarrow k = \frac{1}{49}$$

$$\therefore P(X < 4) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

$$= k + 3k + 5k + 7k = 16k = \frac{16}{49}$$

$$\Rightarrow P(X \geq 5) = P(X=5) + P(X=6)$$

$$= 11k + 13k = 24k = \frac{24}{49}$$

$$\therefore P(3 < X \leq 6)$$

$$= P(X=4) + P(X=5) + P(X=6)$$

$$= 9k + 11k + 13k$$

$$= 33k$$

$$= \frac{33}{49}$$

2) Q A random variable X has the following Probability function:-

$X=x:$	0	1	2	3	4	5	6	7
$P(x):$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

i) Find the value of k

ii) $P(X < 6)$ and

iii) $P(X \geq 6)$

iv) $P(0 < X < 5)$

v) $P(3 < X \leq 6)$

Sol: For the Probability mass function

$$\sum P(x_i) = 1$$

$$\Rightarrow [0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k] = 1$$

$$\Rightarrow 10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k - k - 1 = 0$$

$$\Rightarrow 10k(k+1) - 1(k+1) = 0$$

$$\Rightarrow (k+1)(10k-1) = 0$$

$$\Rightarrow k = -1, \quad k = \frac{1}{10}$$

X

$$\text{So } k = \frac{1}{10}$$

$$\text{ii) } P(X < 6)$$

$$\Rightarrow P(X < 6) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= 0 + k + 2k + 2k + 3k + k^2$$

$$= 8k + k^2$$

$$= \frac{8}{10} + \frac{1}{100} = \frac{800 + 1}{1000}$$

$$= \frac{80 + 1}{100} = \frac{81}{100}$$

$$\text{and iii)} P(X \geq 6) = P(6) + P(7)$$

$$= 2k^2 + 7k^2 + k$$

$$= 9k^2 + k$$

$$= \frac{9}{100} + \frac{1}{10}$$

$$= \frac{9 + 10}{100} = \frac{19}{100}$$

$$\text{iv)} P(0 < X < 5)$$

$$= P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

$$= k + 2k + 2k + 3k$$

$$= 8k$$

$$= 8 \times \frac{1}{10} = \frac{8}{10}$$

$$\text{v)} P(3 < X \leq 6)$$

$$= P(X=4) + P(X=5) + P(X=6)$$

$$= 3k + k^2 + 2k^2$$

$$= 3k + 3k^2$$

$$= \frac{3}{10} + \frac{3}{100} = \frac{30 + 3}{100} = \frac{33}{100}$$